SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I LABHANA PRAVEEN | 192410077 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	6
Experimental Design	30	7
Use of Tools	20	4
Analysis of Results	15	3
Documentation	10	2

Total Marks 22/30

Signature of the Faculty

TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	Basic analysis with limited insights	Poor or incorrect analysis of results

			interpretation		
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



Engineer to Excel

SIMATS ENGINEERING



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Saveetha Institute of Medical And Technical Sciences

(Dedored osDeemed to be University under Section 3 of UGC Ad 1956)

Digital Twin Networks and **the OSI Model**

COI AT3 - TECHNICAL PRESENTATION CSA0710 - COMPUTER NETWORKS

GUIDED BY

Dr. Shankar R,
Professor,
SIMATS
Engineering

PRESENTED BY:

1924260073 LABHANA PRAVEEN

Abstract

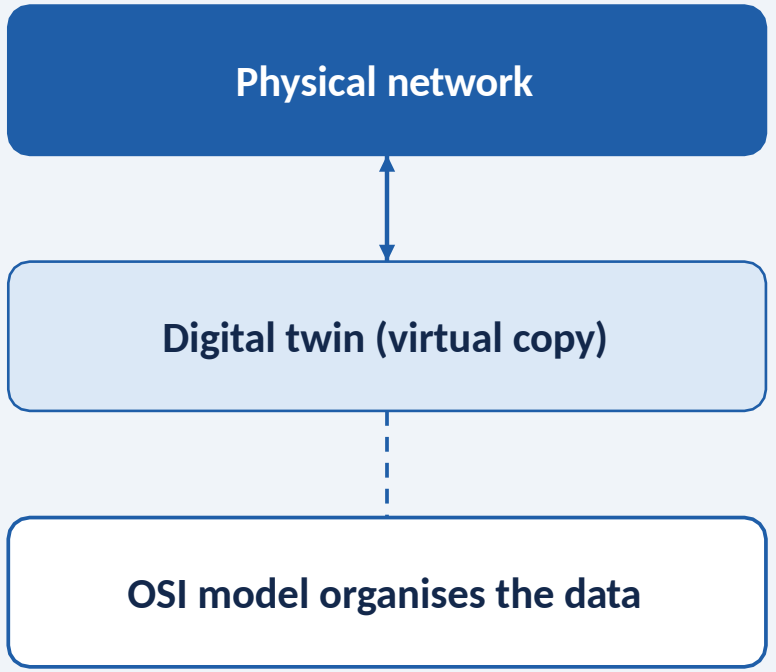
1. A digital twin is a virtual copy of a real network. It is kept up to date with data from the real devices.

2. Engineers can monitor health and study faults on the twin without touching the live network.

3. The OSI model gives a seven-layer map to organise this data and find where a problem starts.

Conceptual study: no live network was tested and no performance results are reported.

The big picture



Introduction

What is a digital twin?

A virtual copy of a physical object or system. Real-time data keeps the copy matching the real one.

What is a digital twin network?

A virtual copy of a real network: its devices, links and traffic. Used to monitor, analyse and test safely.

What is the OSI model?

A seven-layer reference model that explains how data moves in a network, from cables up to applications.

Why use them together?

The twin shows the network state. OSI tells us which layer to check, so faults are found faster.

Problem Statement

Complex networks

Many devices, links, vendors and protocols make the full picture hard to see.

Hard-to-trace failures

One faulty cable or setting can cause many different symptoms.

Slow troubleshooting

Testing fixes on a live network is risky and can cause downtime.

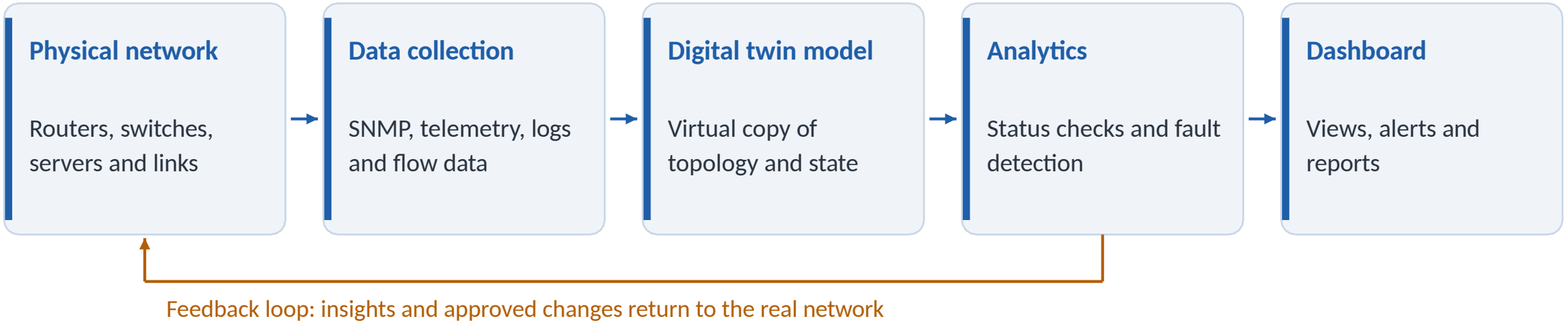
Need: real-time monitoring and early fault detection

Find problems before users notice them.

Objectives

1. Understand digital twin network architecture
2. Study the seven layers of the OSI model
3. Explain how network data can be represented in a digital twin
4. Demonstrate monitoring and fault analysis using a conceptual model

Digital Twin Network Architecture



Real-time data collection

Measures what is happening now: status, traffic and errors from the real devices.

Simulation

Runs what-if tests on the model, such as a link failure, without touching the real network.

OSI Model: Seven Layers

Layer	Main function	Protocol / technology	Real-world example
1 Physical	Sends raw bits as electrical, light or radio signals	Ethernet cable, fibre, Wi-Fi radio	Cable between a PC and a switch
2 Data Link	Moves frames between devices on one link using MAC addresses	Ethernet, MAC addressing	A switch forwarding frames
3 Network	Gives IP addresses and finds the path between networks	IP, ICMP	A router sending traffic to the internet
4 Transport	Delivers data end to end using ports; can be reliable	TCP, UDP	A web page loading over TCP
5 Session	Opens, manages and closes dialogues between applications	RPC, NetBIOS	A login session staying open
6 Presentation	Formats, encodes, compresses and encrypts data	TLS, JPEG, UTF-8	An image shown as JPEG
7 Application	Gives network services to user applications	HTTP, DNS, SMTP	A browser opening a website

Integration: Digital Twin Network and OSI

Layer 1 · Physical	Physical connectivity: link up or down, port status
Layer 2 · Data Link	Packet (frame) delivery on a link: errors and drops
Layer 3 · Network	IP connectivity: reachability, routes and ping
Layer 4 · Transport	Transport connections: TCP sessions, retransmissions
Layers 5 and 6	Session and data-format checks, such as TLS status
Layer 7 · Application	Application performance: web or DNS response time

Not a new layer

The digital twin is not an eighth OSI layer. OSI is only a way to organise the network data that the twin stores and shows.

Proposed System and Working Methodology

1. **Collect network data**
2. **Create or update the twin**
3. **Analyse network status**
4. **Detect anomalies**
5. **Display alerts on dashboard**

Hardware for a prototype

- Lab network: a few routers and switches, real or virtual
- Collector: a PC or Raspberry Pi
- Server or PC to run the digital twin
- Screen for the monitoring dashboard

Software (examples)

- Network emulator, such as GNS3 or EVE-NG
- Data collection: SNMP, syslog or streaming telemetry
- Database and model, such as InfluxDB and NetworkX
- Dashboard and alerts, such as Grafana

Use Case: Network Link Failure

CONCEPTUAL SIMULATION · no live network

Physical network



Uplink cable fault: the Switch to Router link is down

Digital twin



ALERT: uplink down, marked red

1 · What fails

The cable on the Switch uplink is faulty. Users in the lab lose access to the router and the internet.

2 · What we see

In the twin the link turns red. Pings to the router fail (a Layer 3 symptom), but the root cause is at Layer 1.

3 · Alert and fix

Alert: Uplink down on SW-1, Layer 1 (Physical). Check the cable and port, then replace the cable.

Advantages, Applications and Limitations

Advantages

Better visibility
One live view of devices, links and status.

Faster troubleshooting
Find the layer and device at fault quickly.

Predictive analysis
Spot trends and test what-if cases early.

Improved planning
Check capacity and changes before rollout.

Applications

Data centres
Track racks, links and capacity.

Smart campuses
Watch Wi-Fi, switches and campus services.

Industrial IoT
Monitor sensors and factory networks.

Telecom networks
Model large carrier networks and traffic.

Limitations

Data accuracy
Bad or missing data gives a wrong twin.

Implementation cost
Needs tools, skills and setup time.

Security
Collected data and access must be protected.

Synchronization delay
The twin may lag behind the real network.

Conclusion and Future Scope

Summary

- A digital twin network is a live virtual copy of a real network.
- The OSI model organises the data and helps find faults layer by layer.
- The twin is not an OSI layer; the two ideas work together.
- This project shows a conceptual model, not a tested system.

Future scope

AI-based anomaly detection

Learn normal behaviour and flag unusual patterns.

Live network telemetry

Stream real-time data from real devices.

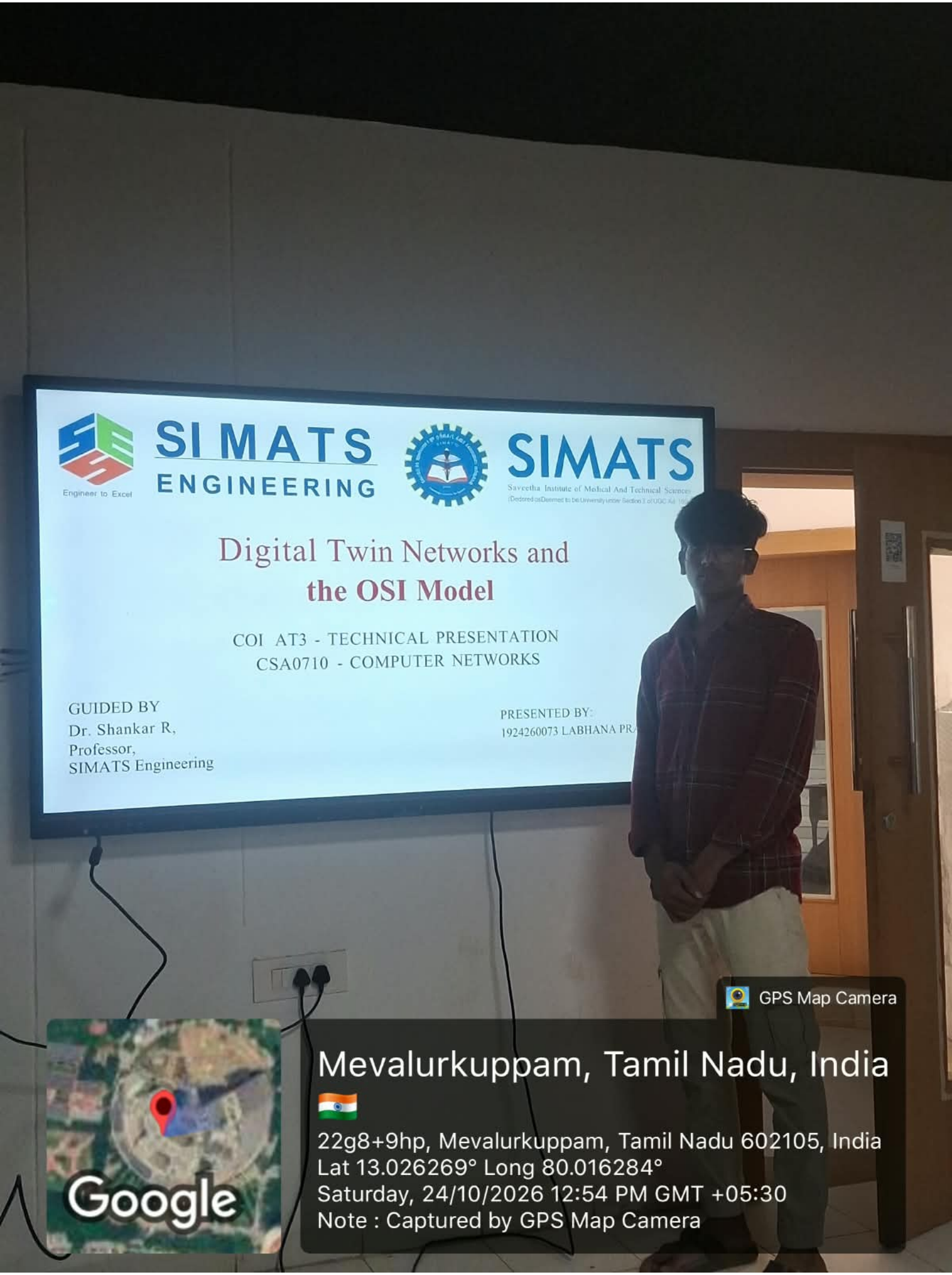
Automated fault diagnosis

Suggest the likely layer and cause automatically.

Thank you · Questions?

References

1. IBM, “What is a digital twin?” ibm.com/topics/digital-twin
2. IBM Developer, “What are digital twins?” developer.ibm.com/articles/what-are-digital-twins
3. Cisco Networking Academy, Introduction to Networks (CCNA): OSI and TCP/IP models module. netacad.com
4. ISO/IEC 7498-1:1994, Information technology – Open Systems Interconnection – Basic Reference Model: The Basic Model.
5. ITU-T Y.3090 (2022), Digital twin network – Requirements and architecture.
6. Y. Wu, K. Zhang and Y. Zhang, “Digital Twin Networks: A Survey,” IEEE Internet of Things Journal, vol. 8, no. 18, 2021.
7. A. Fuller, Z. Fan, C. Day and C. Barlow, “Digital Twin: Enabling Technologies, Challenges and Open Research,” IEEE Access, vol. 8, 2020.


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GPS Map Camera
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Lat 13.026269° Long 80.016284°
Saturday, 24/10/2026 12:54 PM GMT +05:30
Note : Captured by GPS Map Camera
Google